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ORIGINAL PAPER

Thin-Slice Accuracy Depends on the Wisdom of Crowds: Aggregate Judgments Outperform Individual Judgments when Inferring Athletic Performance Based on Nonverbal Behavior

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Abstract
The present paper aims to resolve ambiguous findings on the accuracy or inaccuracy of human judgments based on sparse information by arguing that accurate thin-slice judgments can be understood as driven by the wisdom of the crowd. To test this hypothesis, participants estimated the performance of track and field athletes in different competitions based on thin slices of pre- and post-performance nonverbal behavior. Results suggest that averaged judgments based on thin slices of athletes' nonverbal behavior (NVB) are correlated with actual performance and that averaged judgments are better than individual judgments, thus validating the wisdom of the crowd hypothesis. Taken together, the present paper suggests that understanding thin-slice judgments from a wisdom of the crowd perspective has the potential to resolve the apparent contradiction regarding the accuracy or inaccuracy of human judgments. Furthermore, the paper highlights the potential of averaged thin-slice judgments of naturally occurring NVB as an addition to resource consuming coding procedures.

Keywords Behavioral analysis · Video analysis · Thin slices · Body language

Introduction

Ambady and Rosenthal (1992, p. 269) close their seminal meta-analyses using the so-called *thin slices* paradigm with the statement "the probabilistic expectancies we form of others from very limited information are more accurate than we would expect". This statement apparently confirms the early influential theorizing of Arthur Asch (1946, p. 258) who regarded the human skill of interpersonal accuracy—the skill to accurately assess other

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individuals' emotions, personality, intentions, motives, and thoughts (Schmid Mast & Hall, 2018)—as a "precondition of social life". Subsequent work using the thin-slices paradigm has even argued "that this human capacity is even more remarkable than Asch suggested" (Ambady & Rosenthal, 1992, p. 431) and has led to the fairly undisputed statement in most social psychology textbooks (Ambady et al., 2000) that people are constantly sending out a wealth of information in their expressive behavior that other observers can accurately decode to guide further social interaction. Although we agree with the gist of this statement and do not question that there is a wealth of information to be extracted from thin slices of expressive behavior, we do question how accurately individuals can in fact decode this expressive behavior. This skepticism stems from research on a phenomenon that has been labeled as "the wisdom of the crowd" (Surowiecki, 2004) which shows that individual judgments of humans in a variety of contexts are often less accurate—even of experts in a given field—than the average judgment of a diverse crowd of people. As many studies within the thin-slices literature (Ambady et al., 2000) correlate a criterion (e.g., student teacher evaluations) with an averaged judgment (e.g., averaged teacher effectiveness ratings, Ambady & Rosenthal, 1992) to assess interpersonal accuracy, we argue that the high correlations between the criterion and the thin-slice judgments are oftentimes partly accounted for by "the wisdom of the crowd"—at least if averaging is involved in the judgment variable. To test this assumption, we utilized the field of competitive sports to investigate how judgments of track and field performances based on thin slices of NVB before and after the performance would correlate with the objectively measured performance.

Decoding Nonverbal Behavior and Interpersonal Accuracy

People constantly strive to predict others' behavior, often relying on nonverbal behavior (NVB) as a key source of information. The terms nonverbal communication and NVB are frequently used interchangeably (Hall et al., 2019). In NVB research, individuals are typically categorized as senders (or targets) and receivers (or observers): senders/targets encode large amounts of information through various communication channels (e.g., visual, auditory, olfactory), while receivers/observers decode these NVBs (Matsumoto et al., 2016). The NVBs sent by senders can arise from genetic, biological, developmental, and learned factors. These NVBs may be spontaneous or intentional (posed) and can provide information to receivers, though not all NVBs are necessarily informative (Hall et al., 2019).

Decades of research on nonverbal communication suggest that people cannot avoid communicating nonverbally, as every behavior—or even the lack of behavior—is subject to interpretation (Watzlawick et al., 1967). For instance, an observer might infer that an athlete is focused and concentrated based on their wide, dilated eyes. Such inferences, however, can be accurate or inaccurate. Crucially, the informational value of any NVB or set of NVBs is probabilistic and influenced by additional encoded cues, the characteristics of both the sender and receiver, and contextual factors (Hall et al., 2019). A popular methodology within social psychology providing showcase evidence for the statement that people are well-equipped to encode and accurately decode NVB has been termed *thin-slices paradigm* (Ambady et al., 2000; Ambady & Rosenthal, 1992; Carney et al., 2007). This paradigm involves tests in which participants see and/or hear other people's behavior (usually in short videos or photographs; hence the name *thin slices*) and are asked to infer something about the target person. Stimuli are typically video or audio recorded, after which segments, known as "thin slices," are extracted from these recordings or their transcripts. These experimental stimuli can vary in length, but thin slices are generally very short and no longer than 5 min (Murphy & Hall, 2021). Once extracted, these segments can be analyzed (e.g., coded by trained coders—i.e., descriptive coding, Bente et al., 2008) or rated (e.g., by a group of "naïve" observers—i.e., evaluative coding, Bente et al., 2008) to identify behaviors or characteristics of individuals (senders/targets) involved in the stimuli. Evaluative coding within the thin-slices paradigm are typically presented to a group of observers for judgment of a target's state or traits, if the objective of the research is to evaluate judgment accuracy or interpersonal accuracy. To determine accuracy the observers' inference is subsequently compared or correlated with a criterion variable. For example, a group of observers is asked to judge what kind of emotion a target person is displaying (i.e., judging a state) or certain personality characteristics (i.e., judging a trait) based on observable NVB and these judgments are subsequently compared to self-report measures of emotions, personality or physiological measurement (Hall et al., 2008). A central premise of the paradigm is that extracted slices are representative of the target's behavior during the entire interaction and/or can reveal or predict their internal states, personality, or other social attributes (Murphy & Hall, 2021).

Behavioral researchers often adopt thin-slice techniques for practical reasons, as they help alleviate the burdens of descriptive coding, minimize viewer fatigue, and optimize limited resources like time and patience among both research personnel and participants (Murphy & Hall, 2021). The advantages of thin-slice methods are evident. Given the complexity of human behavior, these methods simplify the challenging task of behavioral measurement, which is widely acknowledged as arduous. As stated in in the opening paragraph, the term "thin slice" was introduced by Ambady and Rosenthal (1992) in a meta-analysis examining the correlations between thin-slice coding or ratings and various outcomes, such as teacher effectiveness ratings, or medical patients' satisfaction. This influential article, cited over 2,880 times as of January 2025, demonstrated the validity and broad applicability of thin-slice methods and solidified thin-slice methodology as a distinct and valuable area of research in a variety of domains (Murphy & Hall, 2021).

Empirical research using the thin-slices paradigm has demonstrated that people can make a broad range of fairly accurate inferences from watching short recordings of the NVB of other people. For example, observer judgments correspond to a high degree with (self-reported) personality characteristics and dispositions of target individuals (Borkenau et al., 2004; Todorov et al., 2005; Willis & Todorov, 2006). Further, observers have been shown to accurately infer mental states of perceived individuals (e.g., Baron-Cohen et al., 1996), what the outcome of a perceived conversation is going to be (Curhan & Pentland, 2007), how student evaluations of a perceived teacher/professor are going to be (Ambady & Gray, 2002), how much self-control capacity a person has at a given moment (Furley et al., 2019), how successful CEOs run a company (Rule & Ambady, 2008), and how successful salespeople are going to be (Ambady et al., 2006).

Pertinent to the present research, a recent series of publications has transferred the thin-slices paradigm to the field of sports as this presents an interesting context to test the thin-slices hypothesis as participant judgments can be compared to objective real-world criteria like the score, race times or jump distances. This thin-slice sport research has concluded that observers can make sport-related judgments and inferences based on their decoding of athletes' NVB (Brimmell et al., 2018; Furley & Memmert, 2021; Furley et al., 2017; Furley & Schweizer, 2014a, b, 2016a, b; Furley et al., 2018). More specifically, research participants could distinguish between athletes leading or trailing during a sport competition (table tennis, basketball, soccer, and handball) based on short video recordings of their NVB (Furley et al., 2023; Furley & Schweizer, 2014a, b, 2016a), between athletes playing at home or away (Furley et al., 2018), they could tell if athletes perceived an upcoming penalty as a challenge or a threat (Brimmell et al., 2018), and how successful an athlete was going to be based on their preparatory behavior in Darts (Furley & Memmert, 2021; Furley et al., 2021).

In summary, the thin-slices literature can be considered to provide converging evidence "that thin-slice judgments can sometimes be surprisingly accurate when accuracy is defined as convergence with independent real-world criteria" (Ambady, 2010, p. 271). This converging evidence from the empirical research has fed into broader theories of nonverbal communication, evolutionary psychology, and social cognition proposing that judgments based on thin slices of behavior are "hardwired" into the human circuitry and occur relatively automatically (DePaulo & Friedman, 1998; Patterson, 1995, 1998, 1999; Tesser & Martin, 1996). Such theorizing is purportedly supported by further empirical evidence showing that thin-slice judgment gets less accurate when observers are asked to deliberate about their judgments (Ambady, 2010). Therefore, there seems to be little doubt about the theoretical tenet that observers of other peoples' NVB often draw accurate inferences from brief observations of this behavior which facilitates adaptive social behavior. This "scientific fact" has been transferred to the broad public by popular science writers like Gladwell (2006).

However, both anecdotal (e.g., Gladwell, 2019) and empirical findings (Naumann et al., 2009; Watson, 1989) indicate that individual observers' judgments of NVB can sometimes be surprisingly inaccurate, although there seems to be conclusive evidence that people on average show high levels of interpersonal accuracy in the thin-slices paradigm. Most likely there are several reasons for this. However, one important point seems to have been overlooked—at least in social psychological textbooks (e.g., Ambady et al., 2000; Fiske et al., 2010)—in explaining why thin-slice judgments "are more accurate than we would expect" (Ambady & Rosenthal, 1992, p. 269): high levels of accuracy in thin-slice studies might partly be a result of the methodology and statistical analyses used within this paradigm: namely aggregating and averaging judgments from a group of independent observers. Several empirical reports from the domain of interpersonal accuracy have compared aggregated vs. non-aggregated observer judgments of a target's state (e.g., pain inference; Ruben & Hall, 2016) or trait (personality inference; Naumann et al., 2009; Watson, 1989). Although it was not the main focus of this research, findings showed that judgment accuracy was higher if the judgments of several observers were aggregated and averaged compared to individual judgments (i.e., one observer rates one target) or averaged individual judgment (i.e., one observer rates multiple targets and these judgments are averaged). Such findings led to the central hypothesis of the present research that high thin-slice judgment accuracy can substantially be reinforced by the "wisdom of the crowd" (Surowiecki, 2004).

The Wisdom of the Crowd

The term wisdom of the crowd has been made popular by the identically named book (Surowiecki, 2004, p. XIV) that entertainingly summarizes evidence showing that "under the right circumstances, groups are remarkably intelligent, and often smarter than the smartest people in them". The book starts of by referring to a story of Sir Galton (1907) who reported that the objectively measured weight of an ox on display at the local fair in Plymouth was almost exactly the same as the median estimate of almost 800 spectators. Galton apparently found this absolutely surprising and irritating as he expected the median guess of the group to be "way off": "After all, a mix of a few very smart people with some mediocre people and a lot of dumb people, and it seems like you'd end up with a dumb answer" (Surowiecki, 2004, p. XII). However, a century of research later has shown that Galton's experience was no exception and compelling evidence has accumulated demonstrating that averaged group judgments are often highly accurate (Larrick et al., 2011, p. 229): "Combining judgments takes individual imperfection and smoothes the edges to isolate the collective's view of the truth. Or to put it more mathematically and mundanely, averaging cancels errors".

According to Soll and Larrick (2006) people have the tendency of seeking, valuing, and relying on expert judgments, although research has shown (e.g., Mannes et al., 2014) that this tendency is not always advisable and can be detrimental: "…chasing the expert is a mistake, and a costly one at that. We should stop hunting and ask the crowd (which of course, includes the geniuses as well as everyone else) instead. Chances are, it knows" (Surowiecki, 2004, p. XV). Obviously, this is not always the best strategy and sometimes individuals are more accurate than crowds (Mannes et al., 2014). The degree to which crowds are more accurate than individuals has been attributed to the factors of knowledge/expertise and diversity within the crowd (Larrick et al., 2011). This means that within crowds there needs to be some degree of knowledge or expertise about the to-be-made judgement and that the crowd need to hold diverse perspectives about the to-be-made judgment for the average judgment to be accurate. This is one of the reasons why we consider it likely that thin-slice judgment accuracy might be an instant of the wisdom of the crowd effect, as people in general are both very knowledgeable about other people due to the amount of social interactions people have from early ages on, but at the same time are also likely diverse in the experiences they have made in these social interactions. Hence, observers in thin-slices paradigms seem to constitute a very wise crowd as they are typically making social judgments (which they are very familiar with), but nevertheless, every individual observer will bring a diverse range of social experiences to the table.

In line with the assumption that the accuracy of thin-slice judgments may at least partly be based on the wisdom of the crowd, we found that out of 44 papers in the meta-analysis by Ambady and Rosenthal (1992) that established the accuracy of thin-slice studies, at least 34 utilize some sort of averaging over individual judgments, which is a necessary condition for wisdom-of-the-crowd effects to occur (see online supplement for the respective information).[1]

Interestingly, the wisdom-of-the-crowd perspective suggests that thin-slice judgments might be even more accurate than estimated by Ambady and Rosenthal (1992): Many thin-slice studies included in the meta-analysis utilize rather small numbers of raters (see online supplement for the respective information). Assuming that every single individual is good

[1] We would like to note that 'accuracy' in the meta-analysis by Ambady and Rosenthal (1992) refers to correlations between thin-slice judgments and different behavioral outcomes (e.g., thin-slice judgments of teachers' personality and students' evaluations of said teachers). In the literature on person perception, just like in the studies reported above, the term 'accuracy' is usually reserved for correlations between two measurements of the same construct (e.g., inferred extraversion and self-reported extraversion). For reasons of conceptual clarity, it seems important to make this difference explicit. Still, from a statistical perspective, it should not matter whether the correlated variables measure the same or different constructs. Statistically, aggregating over several items increases the measurement's reliability, both when the measurement instruments assess the same and when they assess different constructs.

at performing the respective rating task, the number of raters (or judges) should not systematically influence the average individual rating quality. However, from the perspective of the wisdom of the crowd, it does: As wisdom-of-the-crowd effects are based on individual random errors cancelling each other out (Larrick et al., 2011), the average accuracy of wisdom-of-the-crowd judgments should be positively related to the numbers of raters (at least up until a certain point where adding additional raters will not improve accuracy anymore).

The Present Research

The central objective of the present research was to test if thin-slice judgment accuracy can be regarded as an instance of the wisdom of the crowd effect. Therefore, we chose the context of professional sports (i.e., track and field) as it provides a fine-grained objective criterion variable in the form of sprint times or jump distances. In addition, people in general like to watch sports and have experience in making inferences based on athletes' behavior (experience is a perquisite for wisdom of the crowd effects) but also have different levels of experience (diversity prerequisite). Hence, we selected the context of professional track-and-field competitions (100 m spring and long jump) and asked participants to guess sprint times and long jump distances either based on athletes' preperformance or post performance NVB.

The second objective of the present research was to showcase the utility of the thin-slices paradigm for applied fields like sport psychology. Applied fields like sport science face the challenge that artificial laboratory settings often do not transfer to "real world sports" and, therefore, in situ behavioral observation becomes an important methodology to advance understanding of behavior in these real-world situations (Furley, 2019). The sampling and rating methodologies within the thin-slices paradigm (Murphy & Hall, 2021) can arguably be a widely applicable methodological tool for researchers with an applied interest to extract difficult to measure variables via averaged observer ratings in representative samples of a target's real-world behavior (e.g., a persons' feelings or emotions). In this respect, we attempt to utilize the wisdom of the crowd as a way of measuring naturally occurring NVB via evaluative coding (Bente et al., 2008; Furley, 2023).Therefore, we build on theorizing from the field of sport science and psychology suggesting that certain internal states like relaxation, positive affect, confidence are correlated with sports performance (Craft et al., 2003; Jokela & Hanin, 1999) and, of further importance, are correlated with an athletes' NVB (Furley, 2023; Furley et al., 2021; Furley & Memmert, 2021). Hence, an athlete's NVB should differ as a function of performance as measured via the averaged "wisdom of the crowd" judgments in a thin-slices paradigm.

In three different contexts, we asked participants to guess the performance of athletes based on short videos taken either shortly before or after the performance in track and field contexts (see Fig. 1). We asked two groups of participants to guess the performance in 100 m sprint and long jump competitions. The first group of 43 participants was asked to estimate how fast male and female sprinters ran a 100 m sprint based on thin slices of pre-performance video clips shown immediately before the race. The second group of 50 participants was asked to guess the long jump distances of male and female athletes based on thin slices of either pre-performance video clips shown immediately before the run up to the jump or after the jump (without the athlete knowing the measured distance of that jump). A major difference of the long-jump conditions (pre- and post-performance)

Fig. 1 Example images taken from the stimuli used in this research. Top panel: 100 m sprint (pre); middle panel: long jump (pre); bottom panel: long jump (post)

was that it entailed video clips of the same athletes performing either well, average, or poorly. This was implemented to rule out that participants based their judgments primarily on recognizing more or less successful athletes in the 100 m sprint context. Instead, and in line with the applied objective of the present research, we attempted to test if participants' judgments would differ as a function of performance of the same athlete because athletes showed different pre- and post-performance NVB depending on their performance. The rationale for including the pre- and post-performance condition was that we expected stronger correlations between performance and NVB in the post-condition than the pre-condition and therefore it might be easier judging the performance based on post-performance NVB than pre-performance NVB. This allowed us to explore differences in thin-slice judgment accuracy as a function of task difficulty. Arguably, one might expect that the wisdom of the crowd effect might be more pronounced in the difficult condition.

Method

Sample

All datasets (and source code for all statistical analyses) can be found at https://osf.io/8pw2d/?view_only=e71a29f0a6364bbeb7db53b07568d58e. Two samples of participants estimated the performance of the athletes in the present research. The first sample only estimated the performance of male and female 100 m sprinters based on thin slices of pre-performance clips. This sample consisted of 43 participants (26 female and 17 male) with an average age of 26.05 years (*SD* = 8.20). The second sample estimated the performance of male and female long-jumpers based on thin slices of pre-and post-performance clips respectively. This sample consisted of 50 participants (30 female and 20 male) with an average age of 39.94 years (*SD* = 18.49). Sample size considerations were based on previous thin-slice research in sports (Furley & Memmert, 2021; Furley & Schweizer, 2014a) and similar to thin-slices studies in the meta-analysis of Ambady and Rosenthal (1992). Both samples of participants did not have any particular domain-specific knowledge in track and field. Informed consent was obtained online via a button press. The study protocols received ethical approval of the university's ethics board. This study was not preregistered.

Stimuli

We collected three different samples of experimental stimuli.

Pre-performance 100 m sprint. The first sample of stimuli consisted of 49 short video clips and showed world-class male and female 100 m sprinters during their introduction before the finals of the 100 m sprint at the 2015 World Championships in Beijing, the 2016 Olympic Games in Rio, and the 2017 World Championships in London. All video stimuli were sampled to ensure a good frontal visibility of the face and upper body (see top panel of Fig. 1 for an example). The videos were between 3 and 10 s long (*M* = 4.32 s; *SD* 2.11 s for the male videos; *M* = 5.49 s; *SD* 3.24 s for the female videos) and were modified to not reveal any identifying information like the name of the athletes. In addition, all videos did not include any sound.

Pre-performance long jump. The second sample of stimuli consisted of 42 short video clips and showed world-class male (21 clips) and female (21 clips) long jumpers during their preparation of the run up to the jump at the 2019 World Championships in Doha, the 2016 Olympic Games in Rio, the 2020 Olympic Games in Tokyo, and the 2022 European Championships in Munich. The stimulus material consisted of the pre-performance NVB of 7 female and 7 male athletes before a good performance, an average performance, and a poor performance which was calculated in terms of the personal best performance of the respective athlete. Again, all video clips were sampled to ensure good frontal visibility of the face and upper body (see middle panel of Fig. 1 for an example frame taken from the stimulus material). The videos were between 2 and 10 s long (*M* = 5.41 s; *SD* 2.21 s for the male videos; *M* = 6.18 s; *SD* 2.49 s for the female videos) and again were modified to not reveal identifying information or sound.

Post-performance long jump. The third sample of stimuli consisted again of 42 short video clips and showed the post-performance NVB of the same athletes and same jumps as in the second sample. Only this time the clips showed the immediate nonverbal reaction of the athletes after the jump (without them knowing the officially announced jump length yet; see bottom panel of Fig. 1 for an example). The average duration was 4.31 s (*SD* 3.14) sec for the male videos and 5.33 s (*SD* 2.01) sec for the female videos. Otherwise everything was the same as in samples 1 and 2.

Procedure and Measures

We used the software Unipark to administer the online experiment. The software displayed the respective samples of stimuli in different random orders for every participant. This approach helps to ensure that results do not depend on specific sequences of stimuli. After every video clip, participants were asked to estimate the respective performance of the athlete they had just seen the pre- or post-performance NVB of. All videos were presented silently to ensure that ratings were based on NVB and not, for example, crowd noise. Participants gave their ratings by clicking on a continuous scale moving a mouse cursor to a respective point on the scale. Depending on the stimulus set the range of scale encompassed all performances in this stimulus set ranging from the best and worst performance in the respective stimulus set. This was depicted as 10.70 s to 11.80 s for the pre-performance female sprint and 9.70 s to 10.30 s for the pre-performance male sprint. For both the long-jump pre- and post-performance scales it was 624 cm to 730 cm for female athletes and 631 cm to 841 cm for male athletes.

Before commencing the experiment, participants filled out a questionnaire gathering demographic data. After completing the testing procedure, participants were informed about the purpose of the experiment.

Statistical Analysis

In line with the rationale of this work, the statistical analysis aims to shed light on how the effect of aggregating estimations across various test persons (crowd wisdom) influences the correlation between estimated and actual performance of athletes. We do this by tackling three questions: First, does the aggregated opinion of the test persons (i.e. crowd), as to be expected within the thin-slice paradigm, correlate with the actual performance? Second, to which degree is the effect of crowd wisdom responsible for such correlations, i.e. how much weaker are the correlations if considering individual estimations instead of crowd estimations? Third, how does a varying crowd size (i.e. number of test persons) influence the correlation between estimated and actual performance?

To answer the first question, we report Pearson correlation coefficients (and 95% confidence intervals) quantifying the relationship between actual performance and the performance estimations. We consider the test persons' opinion (averaged across the crowd) to include relevant information for the actual performance if we find the correlation coefficients to be positive and deviate significantly from zero. We denote results as significant if p values fall below 0.05 and distinguish between significance levels of 0.05, 0.01 and 0.001. Results are reported separately for each task and we additionally report correlations, when just averaging estimations across all male or female test persons as well as when just averaging across younger (<30 years) or (relatively) older (>=30 years) test persons.

To answer the second question, Pearson correlation coefficients between estimated and actual performances are calculated per test person and then those individual correlations are averaged. Thus, we can infer the effect of the specific crowds in our experiments by comparing the correlation of the crowd (i.e. the correlation of the average performance estimation and the actual performance) to the correlation of individuals (i.e. the average correlation of estimated and actual performance).

We perform and report results of analysis separated by gender (of the athletes) in order to make sure that results do not rely on the trivial effect of test persons being able to correctly predict performance differences between male and female athletes. We excluded one athlete (Christina Williams, 11.8 s) from the 100 m data for further statistical analysis as her running time falls outside the typical range of running times (remaining athletes: 10.92 s$\pm$0.10 s with a range of 10.71 s to 11.09 s). A running time falling more than 0.7 s behind the second worse running time indicates that her running time did not reflect the true performance level, but was influenced by some sort of physical problem like an injury, which is also supported by the video footage of the run. Please note that in the literature on crowd wisdom, the median is often the preferred way to obtain the crowd estimation from the individual estimations (see e.g. Galton [1907]) instead of the mean. Theoretically, it has been argued that the median has advantages over the mean because it is not susceptible to outliers, whereas the median might be a problematic way of aggregation in highly polarized datasets. Given that averaged estimations are predominant in the literature concerning the thin-slice paradigm, we operationalize the crowd estimate as the mean of all individual estimations.

The third question assumes that the effect of crowd wisdom on the results increases with an increasing number of test persons (not to be confused with the number of athletes in the sample). It can be answered by calculating correlations for hypothetical numbers of test persons ranging from 1 person to 100 persons. As one method, we use bootstrapping (Efron & Tibshirani, 1994) to resample the required number of test persons from the actual test persons in our experimental data, using 1000 resamples to calculate each data point. Bootstrapping assumes that the actual probability distribution equals the observed probability distribution and as such test persons can be drawn (with replacement) from the observed distribution. Results, however, are naturally subject to inaccuracies driven by the differences between actual and observed distributions, particularly as, at least for such purposes, crowd sizes in our experiments are small (as is typical in the thin-slices paradigm). As a second method, we use theoretical assumptions on the distribution of actual performances and performance estimations to exactly calculate the effect of crowd size on correlations (see next section). These results are precisely calculable and independent of the specific experimental paradigm, but obviously dependent on the distribution assumptions.

All analyses were performed using R statistical software (v 4.3.2; R Core Team, 2023). Raw datsets in csv format, source code as textfiles and the full R Studio workspace are available online via the open science framework as referenced in the Method section.

Theoretical Model

We denote the observed performance outcomes as $o_i,\ i = 1 \ldots n$ and assume them to be normally distributed with mean μ_o and variance σ_o^2, i.e. $o_i \sim N(\mu_o, \sigma_o^2)$. We denote the estimations of test person $j = 1 \ldots m$ for outcome i as $e_{ij} = o_i + \epsilon_{ij}$, which means that the estimations include an inaccuracy of ϵ_{ij}. We assume these inaccuracies to be normally distributed with mean μ_ϵ and variance σ_ϵ^2, i.e. $\epsilon_{ij} \sim N(\mu_\epsilon, \sigma_\epsilon^2)$. Please note that we assume the inaccuracies to be independent of the observed performances. The estimation of the crowd, denoted as E_i, is simply the average estimation of all persons, which means that $E_i = \frac{1}{m}\sum_{j=1}^{m} e_{ij}$. We are interested in the average correlation $Cor(o_i,\ e_{ij})$ of an individual person's estimations with the observed performance results as well as the correlation $Cor(o_i,\ E_i)$ of the crowd estimation with the observed performance results. From fundamental characteristics of the normal distribution, we can infer the distribution of individual estimation

$$e_{ij} = o_i + \epsilon_{ij} \sim N(\mu_o + \mu_\epsilon, {\sigma_o}^2 + {\sigma_\epsilon}^2)$$

and crowd estimation:

$$\sum_{j=1}^{m} \epsilon_{ij} \sim N(m \cdot \mu_\epsilon, m \cdot {\sigma_\epsilon}^2)$$

$$\rightarrow \ \frac{1}{m}\sum_{j=1}^{m} \epsilon_{ij} \sim N(\mu_\epsilon, \frac{{\sigma_\epsilon}^2}{m})$$

$$\rightarrow \ E_i = \frac{1}{m}\sum_{j=1}^{m} e_{ij} = \frac{1}{m}\sum_{j=1}^{m} (o_i + \epsilon_{ij}) = o_i + \frac{1}{m}\sum_{j=1}^{m} (\epsilon_{ij}) \sim N(\mu_o + \mu_\epsilon, {\sigma_o}^2 + \frac{{\sigma_\epsilon}^2}{m})$$

For the correlation of individuals this yields:

$$Cor\,(o_i,\ e_{ij}) = \frac{Cov(o_i, o_i + \epsilon_{ij})}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + {\sigma_\epsilon}^2}}$$

$$= \frac{Cov\,(o_i, o_i) + Cov(o_i, \epsilon_{ij})}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + {\sigma_\epsilon}^2}}$$

where $Cov\,(o_i, o_i) = {\sigma_o}^2$ and $Cov\,(o_i, \epsilon_{ij}) = 0$ due to the independence of observed performances and errors.

$$\rightarrow \ Cor\,(o_i,\ e_{ij}) = \frac{{\sigma_o}^2}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + {\sigma_\epsilon}^2}} = \sqrt{\frac{{\sigma_o}^2}{{\sigma_o}^2 + {\sigma_\epsilon}^2}}$$

For the correlations of the crowd it yields

$$Cor\,(o_i,\ E_i) = \frac{Cov(o_i, \frac{1}{m}\sum_{j=1}^{m} (o_i + \epsilon_{ij}))}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + \frac{{\sigma_\epsilon}^2}{m}}} = \frac{\frac{1}{m} Cov(o_i, \sum_{j=1}^{m} (o_i + \epsilon_{ij}))}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + \frac{{\sigma_\epsilon}^2}{m}}}$$

$$= \frac{\frac{1}{m} \cdot m \cdot Cov(o_i, o_i + \epsilon_{ij})}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + \frac{{\sigma_\epsilon}^2}{m}}} = \frac{Cov(o_i, o_i + \epsilon_{ij})}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + \frac{{\sigma_\epsilon}^2}{m}}}$$

$$= \frac{{\sigma_o}^2}{\sqrt{{\sigma_o}^2}\sqrt{{\sigma_o}^2 + \frac{{\sigma_\epsilon}^2}{m}}} = \sqrt{\frac{{\sigma_o}^2}{{\sigma_o}^2 + \frac{{\sigma_\epsilon}^2}{m}}}$$

Please note that for a crowd of one person ($m = 1$), the latter correlation obviously equals the former correlation. Please also note that averaging across the individual correlations does not change its theoretical value.

Results

Correlations between the aggregated estimates of test persons and the actual performances range between 0.44 for male athletes in 100 m and 0.94 for female athletes in the long jump post condition and reach significance across all genders and conditions. Except for male athletes in 100 m, all correlations would be considered large effect sizes with reference to Cohen (1988) and we would conclude that humans are very good in estimating the performances based on thin slices. Results across both genders, all three conditions (100 m, long jump pre-condition, long jump post condition) and for various subgroups of the test persons are summarized in Table 1. In all three conditions, correlations are higher for estimations made for female athletes (not to be confused with the ones made by female test persons) than those made for male athletes. The correlations are higher for predicting long jump distances before the attempt than for predicting 100 m running time and non-surprisingly are higher for predicting long jump distances in the post than in the pre-condition. As illustrated by the confidence intervals and caused by the limited number of performances in the data, correlations are subject to substantial noise. As such, it is difficult to draw conclusions on differences between correlations of subgroups. All in all, correlations appear to be very homogeneous across subgroups of the test persons and interestingly, the only groups closely failing to reach significance are male test persons for female athletes in the 100 m condition

Table 1 Pearson correlation coefficients between the averaged estimation of the crowd and the actual performances

Condition	Test persons	Male athletes			Female athletes		
		Correlation	CI	p-value	Correlation	CI	p-value
100 m	**All**	**0.44**	**[0.06, 0.71]**	**<0.05**	**0.53**	**[0.16, 0.78]**	**<0.01**
	Male	0.41	[0.02, 0.69]	<0.05	0.37	[-0.04,0.68]	0.078
	Female	0.46	[0.08, 0.72]	<0.05	0.66	[0.34, 0.84]	<0.001
	Older	0.48	[0.20, 0.73]	<0.05	0.48	[0.09,0.75]	<0.05
	Younger	0.43	[0.04, 0.70]	<0.05	0.49	[0.10,0.75]	<0.05
Long Jump Pre	**All**	**0.57**	**[0.19, 0.81]**	**<0.01**	**0.78**	**[0.53, 0.91]**	**<0.001**
	Male	0.71	[0.39, 0.87]	<0.001	0.83	[0.63, 0.93]	<0.001
	Female	0.42	[-0.02, 0.72]	0.060	0.70	[0.38, 0.87]	<0.001
	Older	0.59	[0.21, 0.81]	<0.05	0.76	[0.49, 0.90]	<0.001
	Younger	0.48	[0.06, 0.76]	<0.05	0.73	[0.44, 0.88]	<0.001
Long Jump Post	**All**	**0.74**	**[0.46, 0.89]**	**<0.001**	**0.94**	**[0.85, 0.97]**	**<0.001**
	Male	0.77	[0.51, 0.90]	<0.001	0.93	[0.83, 0.97]	<0.001
	Female	0.72	[0.42, 0.88]	<0.001	0.93	[0.83, 0.97]	<0.001
	Older	0.74	[0.45, 0.89]	<0.001	0.93	[0.83, 0.97]	<0.001
	Younger	0.74	[0.46, 0.89]	<0.001	0.89	[0.75, 0.96]	<0.001

Note. Values are calculated separately by athlete gender and experimental condition. Crowd estimations are reported for all test persons and additionally for specific subgroups of the test persons (female, male, younger than 30 years, 30 years or older). For reasons of clarity, we refrain from repeatedly reporting degrees of freedom (df=23 for 100 m male, df=21 for 100 m women, df=19 for both jump conditions and genders) in the Table and female test persons for male athletes in the long jump pre condition. Thus, it might be interesting to evaluate whether persons are better in estimating athletes of their own gender. Finding a clear answer to this question, however, is not possible based on the present dataset and also does not match the main focus of the present study.

Table 2 summarises results for the individual correlations of test persons' estimations with the actual performances. In all cases, the average values for the individual correlations, as to be expected, are considerably smaller than the correlations of the crowd. The differences between averaged individual and crowd correlations provide an assessment for the influence of crowd wisdom in the specific case of the present datasets. It becomes apparent that the ability of the crowd to estimate the performances based on thin slices, to a high degree, is dependent on the aggregation across individuals.

While the above results are based on the particular number of test persons in the specific paradigms used, the effect of the crowd aggregation on correlation is illustrated in Figs. 2 and 3 for crowd sizes ranging from 1 to 100 persons. The former is based on bootstrapping methods and thus still reflects the data from the specific paradigms. The latter is based on the theoretical model and thus independent of any specific paradigm, but dependent on the distribution assumptions.

Results of both methods are widely consistent and confirm the expected crowd wisdom effect leading to a continuous increase in correlations if adding persons to the crowd. It is striking that even small groups of 5 or 10 persons are often sufficient to achieve a massive increase in correlation. Please note that bootstrapping with the given data comes with the problem of having a limited number of test persons to resample from. As such, the distribution from the data might be a quite inaccurate representation of the underlying true distribution. Particularly, resampling a large number of persons unavoidably includes doubled observations. This is an important reason why the additional theoretical model is needed and a potential reason why the correlations based on bootstrapping seem to have a ceiling effect not observable for the theoretical model. A further potential reason for this is that the strict assumptions of the model (e.g. independence of estimation errors from performance values or independence of individual estimations) are potentially not fulfilled in real-world data.

Discussion

The present research addressed two goals. On a theoretical level, we tested the assumption that thin-slice judgment accuracy can be regarded as an instance of the wisdom of the crowd phenomenon. The second, applied objective was to illustrate that averaged judgments in the

Table 2 Information on the Pearson correlation coefficients between the individual estimation of test persons and the actual performances

Condition	Male athletes					Female athletes				
	Mean	Median	SD	Min	Max	Mean	Median	SD	Min	Max
100 m	**0.26**	0.24	0.21	−0.09	0.70	**0.18**	0.20	0.22	−0.24	0.57
Long Jump Pre	**0.27**	0.24	0.23	−0.21	0.94	**0.36**	0.39	0.23	−0.13	0.78
Long Jump Post	**0.61**	0.63	0.14	0.26	0.96	**0.64**	0.64	0.12	0.33	0.91

Note. For each athlete gender and all experimental conditions, the mean, median, standard deviation (SD), as well as maximum (max) and minimum (min) of the individual correlations is reported. Again, degrees of freedom (df=23 for 100 m male, df=21 for 100 m women, df=19 for both jump conditions and genders) are not repeated in the Table

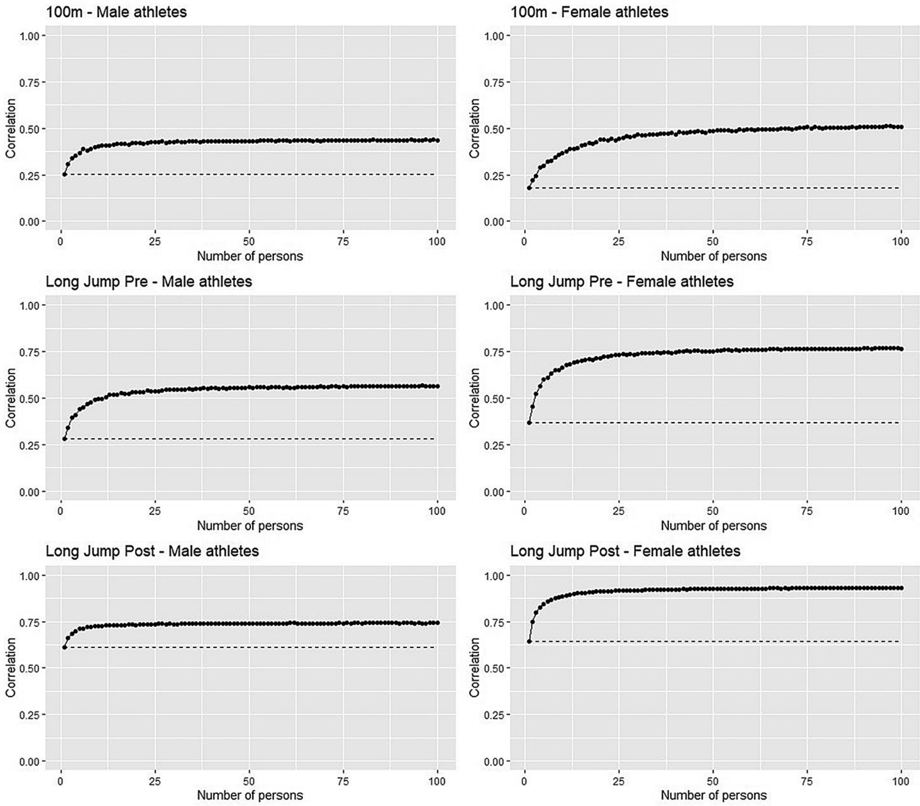

Fig. 2 Influence of crowd size on correlations based on the results of bootstrapping. The number of persons is illustrated on the x-axis and the correlation between crowd estimation and observed performances is illustrated on the y-axis. Plots are based on 1000 resamples per data point and are separated by the three experimental conditions as well as by gender

thin-slices paradigm can be utilized as a useful dependent variable in the context of naturalistic behavioral observations. To these ends we utilized the field of competitive sports to investigate how judgments of track and field performances based on thin slices of NVB before and after the performance would correlate with the objectively measured performance. The pattern of results clearly shows that averaged judgments correlated significantly with actual performance. Importantly, further analyses revealed that the "crowd" (averaged judgments) outperformed individuals on the majority of judgments and therefore support the central hypothesis that thin-slice judgment accuracy is, at least partly, attributable to the wisdom of the crowd phenomenon. Regarding the second objective, the present research has heuristic value as it shows that averaged judgments can be rather accurate and therefore can be a useful dependent variable in real world behavior observations (a point we return to later in the discussion).

Regarding the central objective of the present research, we do not claim that we (re) discovered the wisdom of the crowd effect, or more generally the well-known effect that in many instances aggregate judgments outperform single judgments. Rather, we aim to correct a somewhat overoptimistic impression that may have arisen from previous studies

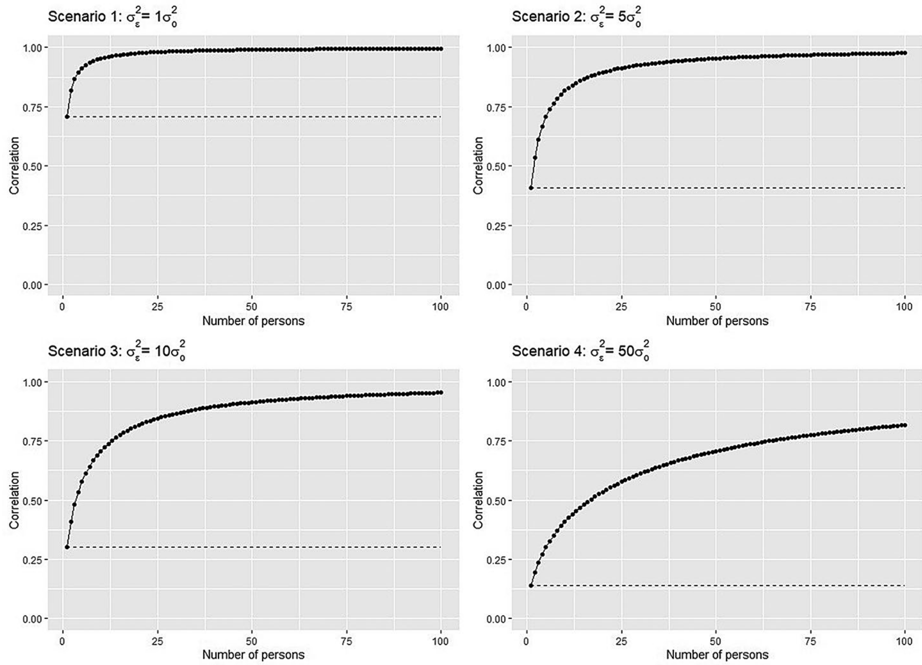

Fig. 3 Influence of crowd size on correlations based on the theoretical model. The number of persons is illustrated on the x-axis and the correlation between crowd estimation and observed performances is illustrated on the y-axis. Plots are based on four scenarios where the variance of the estimation errors increases in relation to the variance of the actual performances

on person perception, thin slices and NVB: namely the impression that humans can make accurate inferences regarding other persons' states, traits or characteristics based on a single exposure to other persons' behavior. Sometimes, of course, humans are able to do this. More often, however, they are able to infer other persons' characteristics not based on a *single* brief exposure of the other person's expressive behavior, but on the aggregate of *multiple*, brief exposures. Still, the single exposure itself can be very brief.

This important theoretical tenet helps to clarify a seeming paradox within popular science literature on person perception and nonverbal communication by best-selling author Malcom Gladwell. Although Gladwell enthusiastically wrote about how accurately an observer could decode NVB in the "blink" of an eye in his best-selling book *Blink* (Gladwell, 2006; with over 5.790 citations as of January 2025), he (Gladwell, 2019) somewhat surprisingly changed his mind about this in a more recent publication ("Talking to Strangers"). By presenting high profile historical anecdotes of how individuals failed at accurately interpreting the NVB of other individuals (after short behavioral observations that arguably resemble "thin-slice studies"), Gladwell initiates his line of argumentation. In one such high profile example, Gladwell outlines a meeting between British prime minister Neville Chamberlain and German chancellor Adolf Hitler in September, 1938 at the brink of World War 2. After this meeting Chamberlain is convinced that Hitler has no intentions of war and therefore announces to a cheering crowd "I believe it is peace for our time" (Gladwell, 2019, p. 45). Since Hitler soon thereafter invaded Czechoslovakia and Poland and started World War 2, Gladwell (2019, p. 45) took this vivid example together with other historical anecdotes to illustrate the central argument of the book that individuals often have difficulties in accurately interpreting a strangers' observable behavior:

> "We have, in other words, CIA officers who cannot make sense of their spies, judges who cannot make sense of their defendants, and prime ministers who cannot make sense of their adversaries. We have people struggling with their first impression of a stranger. We have people struggling when they have months to understand a stranger. We have people struggling when they meet with someone only once, and people struggling when they return to a stranger again and again. They struggle with assessing a stranger's honesty, they struggle with a stranger's character. They struggle with a stranger's intent. It's a mess".

Together with previous findings from the interpersonal accuracy literature (Naumann et al., 2009; Ruben & Hall, 2016; Watson, 1989), the present results suggest that high levels of accuracy in thin-slice studies (i.e., large correlations between aggregated judgments and a criterion) are partly a result of the methodology and statistical analyses used within this paradigm and can therefore be regarded as a "wisdom of the crowd" effect (Surowiecki, 2004). In this regard, it seems important to emphasize that this does not mean that inferences based on aggregates of thin slices are some sort of statistical artefact. Humans are able to make systematic inferences regarding other persons' characteristics based on the aggregate of several single expressions precisely *because* humans do convey meaningful information via very brief instances of their expressive NVB. However, this signal is highly probabilistic, i.e., it is rather small as compared to the associated noise and thus only works on the aggregate.

Thus, instead of writing "our results suggest that perceivers can reliably or accurately interpret brief slices of human NVB", it would be more correct to write "our results suggest that perceivers can *on average* reliably or accurately interpret brief slices of human NVB". Given that in most social situations perceivers do not encounter single brief instances of expressive behavior but instead multiple ones, it becomes obvious why interpreting aggregates of brief instances of expressive behavior may still be highly adaptive. For instance, when athletes enter the pitch, they do not see a single nonverbal expression by their opponents, but rather a stream of NVBs, each of which may only last for the fraction of a second, but which taken together may be sufficient to allow for an informed assessment of their opponents' states.

This brings us to the applied objective of the present research. The finding that averaged thin-slice performance estimates significantly correlated with objectively measured track-and-field performances implies that individual people within the groups of participants were able to pick up valid cues (in regards to performance) from the behavioral observations of male and female sprinters and long-jumpers. This further implies that track-and-field athletes likely sent out cues that were reflective of their momentary inner states like relaxation, positive affect, confidence which are correlated with sports performance (Craft et al., 2003; Jokela & Hanin, 1999). The higher correlations for the post-performance stimuli somewhat support this reasoning as an athlete has received feedback about his/her performance, which in turn might lead to stronger feelings/emotions compared to the pre-performance condition and therefore it might be easier judging the performance based on post-performance NVB than pre-performance NVB. Taken together, we consider averaged thin-slice judgments a useful heuristic measure of NVB that is related to difficult to measure states like emotions in a naturalistic setting. In this respect, evaluative coding (Bente et al., 2008) via averaged thin-slices judgments can be a useful measure in NVB research—in addition or as an alternative—to resource consuming facial (Ekman & Friesen, 1978; Ekman et al., 1971) or bodily coding procedures (Furley & Roth, 2021). Typically, descriptive coding studies serve different objectives than evaluative coding studies (Bente et al., 2008) by attempting to illuminate specific behaviors within Brunswik's (1956) lens model approach, whereas evaluative coding (e.g., emotion or personality ratings within the thin-slices paradigm) attempts to find out how specific behaviors are interpreted by observers. Hence, facial and bodily coding procedures provide detailed descriptions of facial and bodily actions, while averaged judgments in the thin-slices paradigm can be regarded as an evaluative procedure (Furley, 2021) that provides a more holistic measure of a person's demeanor in the sense of the famous Gestalt psychologist saying that "the whole is more than the sum of its parts". Regarding the measurement of NVB this means that via the wisdom of the crowd effect, averaged judgments within the thin-slices paradigm can provide a valid holistic measure of a person's demeanor. In this regard, future NVB research might want to contrast descriptive coding procedures by e.g. expert FACS coders assessing emotional states with averaged evaluative ratings by a group of untrained observers. According to the introduced wisdom of the crowd theorizing it seems a viable hypothesis that aggregated crowd ratings of emotional states in a given situation might be more accurate that coded emotional states by expert FACS coders (Mannes et al., 2014; Soll & Larrick (2006): "…chasing the expert is a mistake, and a costly one at that. We should stop hunting and ask the crowd (which of course, includes the geniuses as well as everyone else) instead. Chances are, it knows" (Surowiecki, 2004, p. XV).

Previous thin-slice sport studies support the reasoning that averaged judgments can be considered a valid measure of a person's demeanor or NVB in a naturalistic context as averaged judgments of athletes' NVB distinguished between athletes leading or trailing during a sport competition (table tennis, basketball, soccer, and handball; Furley & Schweizer, 2014a, b, 2016a), between athletes playing at home or away (Furley et al., 2018), between athletes perceiving an upcoming penalty as a challenge or a threat (Brimmell et al., 2018), and between successful and unsuccessful performances based on their pre-performance NVB in Darts (Furley & Memmert, 2021; Furley et al., 2021). The present findings add to this line of research by providing evidence that world-class track-and-field athletes send nonverbal cues before and after their performances that observers can pick up to infer the performances of the respective athletes—at least if the performance inferences are averaged across the group of observers. Not surprisingly, these performance inferences based on thin slices of athletes' NVB were more accurate when the videos showed an athletes' post-performance NVB instead of their pre-performance NVB. This was likely due to the fact that an athletes' NVB showed more valid cues that were correlated to the performance in the post-performance than the pre-performance condition due to feedback about the performance in the post-condition. In the post-condition the athlete had much more information about the performance (even if they hadn't been informed about the long-jump distance yet) and this likely showed in the athletes NVB: e.g., being happy or disappointed about the performance.

The present research had some notable strengths and limitations. We consider the careful selection and random presentation of the video clips to be a major strength of the present work. Furthermore, the selection of the long jump stimuli (i.e., a good, an average and a poor performance per athlete) allows to rule out an important alternative explanation, namely that perceivers based their judgments on the recognition of certain athletes. Regarding the limitations of the present study, despite the careful selection of video stimuli across a range of competitions and performances, an even more heterogenous sample of stimuli might have been desirable. Furthermore, the sample sizes in the present study are rather small, although they were based on previous thin-slice sport studies and are notably larger than most “crowds” in the thin-slice studies analyzed in Ambady and Rosenthal’s (1992) seminal metanalysis (see online supplement). However, as discussed briefly above, the advantage of the wisdom of the crowd over individual performances should (at least up to a certain point) be positively related to the size of the crowd as we show in Figs. 2 and 3. Thus, in all likelihood, with a larger sample size (i.e. crowd size), averaged judgments would have outperformed individual judgments even more strongly. Additionally, participants in the present study represent a rather homogenous group regarding factors such as native language, nationality, and ethnic and cultural background. In conclusion, generalizability of the present findings to different settings and people is uncertain. Finally, it should be noted that, as common in the thin-slices literature, we use the term accuracy to refer to high correlations between estimations and actual performances. Statistically, a high correlation does not necessarily imply accuracy in the sense of small differences between each estimation and the actual performance. For example, a high correlation between estimates and actual performances might result when the relative positions of the estimated performances (i.e., their rankings) correspond to the relative positions of the actual performances, although the single differences between estimated and actual performances are substantial.

More generally, in psychological studies judgments are often described as being accurate based on a criterion for accuracy that does not require the majority of individual judgments to be close to the estimated values. For example, accuracy may be inferred when a correlation is significantly different from zero or when the mean of the judgments differs significantly from the mean you would expect by guessing alone. In such instances, results suggest that judges or perceivers are picking up and using some signal when making their judgments, and respective conclusions with regard to theory may be drawn (e.g., the results suggest that athletes nonverbally send some signal associated with performance, and humans can on average interpret these signals). However, readers should not leap to any conclusions regarding the accuracy of single judgments (e.g., every time a perceiver sees an athlete before their performance the perceiver can accurately predict what exactly the next performance is going to be like).

In conclusion, results show that in line with the idea of crowd wisdom, individual estimations were demonstrably worse than those of the crowd. It should be noted that the main rationale behind this work is not to show that crowd wisdom exists nor to demonstrate that it can be found in the very specific experimental conditions of this study. The existence of crowd wisdom as a statistical phenomenon is undisputed and has even been shown to be robust against factors like biased individual estimations or violations of independence on individual estimations (Davis-Stober et al., 2014). In this respect, it seems very unlikely to find a dataset, where the idea of crowd wisdom is negated. This is particularly true as within this study the datasets consist of estimations given in absence of any social interaction and analyses the crowd estimations in comparison to all individuals not considering the role of individuals with a potential specific expertise in the task (i.e. experts). Instead, the rationale of this study is to create awareness that crowd wisdom may play a substantial role in the apparent ability of humans to identify relevant information in thin slices of information like the visual stimuli of athletes. A more careful consideration of differences between individual and crowd estimations is needed to better understand whether individual humans actually are successful in a specific task or only groups of humans are.

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Data Availability Raw datasets as well as source code for statistical analyses can be found here: https://osf.io/8pw2d/?view_only=e71a29f0a6364bbeb7db53b07568d58e.

Declarations

Research Disclosure The authors declare that (1) (a) the total number of excluded observations and (b) the reasons for making these exclusions have been reported in this manuscript; (2) that all independent variables or manipulations, whether successful or failed, have been reported in the manuscript; (3) that all dependent variables or measures that were analyzed for this article's target research question have been reported in in the manuscript/online supplements; (4) that (a) how sample size was determined and (b) our data-collection stopping rule have been reported in online supplements to this manuscript.

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